Comprehensive Critical Care MCQ Question Bank

1. Types of Shock

1. Which hemodynamic monitoring tool is specifically noted in the source for providing accurate assessment of cardiac function and output in critically ill patients?

A. Pulmonary artery catheter

B. Pulse oximetry

C. Nasogastric tube

D. Arterial Blood Gas (ABG) analysis

Answer: A

2. In the context of intrinsic lung and airway diseases, which condition is identified as a cardiovascular cause of Acute Respiratory Failure (ARF)?

A. Pulmonary emphysema

B. Chronic bronchitis

C. Cardio pulmonary edema

D. Large airway obstruction

Answer: C

3. According to the management principles of critically ill patients, what is the primary goal of optimizing cardiac function?

A. To decrease metabolic oxygen requirements

B. To maintain adequate cardiac output

C. To prevent alveolar consolidation

D. To induce respiratory alkalosis

Answer: B

2. ICU Monitoring: Vital Signs and Ranges

4. To ensure adequate tissue oxygenation while avoiding toxicity, what is the targeted arterial hemoglobin oxygen saturation (SaO2) goal?

A. Greater than 85%

B. Exactly 100%

C. Greater than 90%

D. Between 60% and 80%

Answer: C

5. Type I Acute Hypoxemic Respiratory Failure is clinically defined by which of the following arterial blood gas values?

A. PaO2 of 80 mmHg or more

B. PaO2 of 50 mmHg or less

C. PaCO2 of 60 mmHg or more

D. pH of 7.50 or more

Answer: B

6. Which diagnostic study is described as the only method capable of confirming a diagnosis of ARF by determining exact gas levels and pH?

A. Chest radiography

B. Sputum culture

C. Pulmonary function tests (PFT)

D. Arterial Blood Gas (ABG)

Answer: D

7. In the periodic assessment of therapy response, which non-invasive tool is used for the continuous monitoring of arterial oxygen saturation?

A. Pulmonary artery catheter

B. Pulse oximetry

C. Capnography (EtCO2)

D. Bronchoscopy

Answer: B

3. VAP Bundle

8. Based on the strategies for treating underlying causes, which intervention is a priority for preventing infectious complications in ventilated patients?

A. Administration of high-dose corticosteroids

B. Prevention and treatment of respiratory tract infections

C. Routine use of sedative and narcotic drugs

D. Maintaining a state of alveolar hypoventilation

Answer: B

9. To prevent mechanical complications related to the airway, clinicians should focus on identifying and treating which of the following?

A. Night sweats

B. Potential airway obstruction

C. Morning headache

D. Peripheral nerve disorders

Answer: B

10. Which management step is essential for clearing foreign bodies or secretions to maintain airway patency?

A. Pursed lip breathing

B. Clearing foreign bodies and secretions

C. Reducing positive airway pressure

D. Increasing sedative dosages

Answer: B

11. Management aimed at preventing complications of ARF includes the assessment and treatment of which cardiac-related trigger?

A. Chronic bronchitis

B. Congestive heart failure

C. Sleep apnea syndrome

D. Pleural effusion

Answer: B

4. Indications for Mechanical Ventilation

12. At what physiological threshold does the source indicate a patient typically requires the initiation of invasive mechanical ventilation?

A. Blood pH less than 7.25

B. Blood pH greater than 7.45

C. $PaCO_2$ less than 35 mmHg

D. PaO_2 greater than 95 mmHg

Answer: A

13. For patients requiring volume ventilation due to severe respiratory distress, which specific ventilator mode is recommended?

A. Pressure Support

B. Assist/Control mode

C. Continuous Positive Airway Pressure (CPAP)

D. Pursed lip breathing

Answer: B

14. Type II (Ventilatory) Respiratory Failure is characterized by which of the following profiles?

A. $PaO_2 > 60$ mmHg and $pH > 7.40$

B. $PaCO_2 > 50$ mmHg and $pH \leq 7.30$

C. $PaCO_2 < 40$ mmHg and $pH > 7.45$

D. Normal $PaCO_2$ with low SaO_2

Answer: B

5. Causes of Respiratory Failure

15. Which of the following is categorized as a "Parenchymal Disease" leading to intrinsic respiratory failure?

A. Asthma

B. Severe pneumonia

C. Foreign body obstruction

D. Chronic bronchitis

Answer: B

16. Flail chest and traumatic injury to the chest wall are classified under which category of respiratory failure causes?

A. Central Nervous System disorders

B. Diseases of the Pleura and the Chest Wall

C. Peripheral nerve disorders

D. Bronchial diseases

Answer: B

17. Which of the following is an "Extrapulmonary Disorder" caused by Central Nervous System dysfunction?

A. Guillain-Barré syndrome

B. Cerebrovascular accident (Stroke)

C. Poliomyelitis

D. Spinal cord trauma

Answer: B

18. Intrapulmonary shunting, a major cause of hypoxemia, is often the result of which underlying condition?

A. Sleep apnea syndrome

B. Alveolar collapse (atelectasis)

C. Chest wall trauma

D. Sedative overdose

Answer: B

6. Glasgow Coma Scale (GCS) Scoring

19. While the GCS is a standard tool, clinical assessment of "Cerebral physical findings" in Type I RF focuses on identifying which symptoms?

A. Muscle tremors and tremors

B. Restlessness, confusion, and coma

C. Pursed lip breathing and cyanosis

D. Hypotension and arrhythmia

Answer: B

20. In Type II Hypercapnic failure, which specific cerebral finding is noted as a result of carbon dioxide retention?

A. Cyanosis

B. Morning headache and disorientation

C. Prolonged expiration

D. Tachycardia

Answer: B

21. A patient presenting in a "coma" during a respiratory crisis may be experiencing which of the following?

A. Successful gas exchange

B. Severe hypoxemia or hypercapnia

C. Effective compensation

D. Resolution of underlying cause

Answer: B

7. Acute Coronary Syndrome (ACS) Occlusion Types

22. According to the list of intrinsic causes, which cardiovascular emergency can lead directly to acute respiratory failure?

A. Pulmonary emphysema

B. Cardio pulmonary edema

C. Pneumothorax

D. Spinal cord trauma

Answer: B

23. Hemodynamic monitoring in patients with suspected cardiac-induced respiratory failure involves the use of which diagnostic tool to evaluate heart function?

A. Sputum culture

B. ECG and Echocardiography

C. Pulmonary function tests

D. Chest radiography

Answer: B

24. Maintaining adequate cardiac output is a pillar of management for patients whose respiratory failure is triggered by:

A. Large airway obstruction

B. Cardiovascular disease

C. Pleural effusion

D. Chronic bronchitis

Answer: B

8. Vagotonic Maneuvers

25. Patients with Hypoxemia (Type I RF) frequently manifest which of the following cardiac physical findings?

A. Bradycardia

B. Tachycardia and arrhythmia

C. Normal sinus rhythm

D. Absence of pulse

Answer: B

26. To manage dysrhythmias in the critical care setting, the clinician must first address which of the following precipitating factors?

A. Overfeeding

B. Hypoxemia, acidosis, and electrolyte imbalances

C. Immobility

D. Daily weight measurements

Answer: B

27. Which clinical sign is an indicator of fatigue and impending respiratory arrest in the hypoxemic patient?

A. Cyanosis

B. Inability to speak

C. Hypertension

D. Use of accessory muscles

Answer: B

9. Nursing Interventions for Intracranial Pressure (ICP)

28. Head trauma and sedative overdose are examples of disorders affecting which system?

A. Peripheral nervous system

B. Central Nervous System (CNS)

C. Pleural cavity

D. Bronchial tree

Answer: B

29. Which complication of ARF specifically affects the brain due to inadequate oxygen delivery and acidosis?

A. Venous thromboembolism

B. Ischemic anoxic encephalopathy

C. Abdominal distension

D. Alveolar consolidation

Answer: B

30. In patients with CNS-induced respiratory failure, what must the nurse assess regarding thoracic movement?

A. Sputum color

B. Limitation of thoracic wall movement and cough effectiveness

C. Fluid retention in the extremities

D. Presence of sinus problems

Answer: B

10. Complications of DKA

31. Life-threatening dysrhythmias in the critically ill can be precipitated by which metabolic state?

A. Respiratory alkalosis

B. Acidosis and electrolyte imbalances

C. Hypernatremia

D. Normal blood pH

Answer: B

32. Ischemic anoxic encephalopathy is a severe complication resulting from a combination of:

A. Hyperkalemia and nutrition

B. Hypoxemia, hypercapnia, and acidosis

C. Normal PaO2 and PaCO2

D. Low protein intake

Answer: B

33. Which systemic complication is associated with venous stasis and immobility in the ICU?

A. Ischemic encephalopathy

B. Venous thromboembolism (VTE)

C. Alveolar hypoventilation

D. Bronchospasm

Answer: B

11. Treatment of DKA

34. What is a primary goal of fluid restoration in the management of the critically ill?

A. To induce rapid weight gain

B. To prevent excessive intravenous fluid or poor intake

C. To maximize pulmonary artery pressure

D. To eliminate the need for ABG monitoring

Answer: B

35. Which specific electrolyte imbalances must be prevented and treated promptly to maintain stability?

A. Hypercalcemia and hypernatremia

B. Hypokalemia and hypophosphatemia

C. Hypomagnesemia and hyperchloremia

D. Normal electrolyte levels

Answer: B

36. To monitor fluid and electrolyte balance effectively, which nursing action is required daily?

A. Sputum volume measurement

B. Daily body weight measurement

C. Assessment of night sweats

D. Chest radiography

Answer: B

37. Nutrition support in the intensive care setting should be initiated while strictly avoiding which of the following?

A. Supplemental oxygen

B. Overfeeding

C. Bronchodilators

D. Enteral routes

Answer: B

12. Cerebral Stroke: Types and Target Blood Pressure

38. A cerebrovascular accident (stroke) is classified as which type of cause for respiratory failure?

A. Intrinsic airway disease

B. Extrapulmonary CNS disorder

C. Disease of the pleura

D. Parenchymal disease

Answer: B

39. Physical findings of cerebral distress in a patient with acute hypoxemia (Type I) include:

A. Muscle tremors

B. Restlessness, confusion, and coma

C. Pursed lip breathing

D. Fatigue

Answer: B

40. Patients with Type II (Hypercapnic) failure often present with which specific neurological symptom upon awakening?

A. Blurred vision

B. Headache on awakening

C. Sudden hearing loss

D. Hyper-reflexia

Answer: B

41. Which hemodynamic finding is associated with the cardiovascular response to hypoxemia?

A. Hypotension only

B. Hypertension or hypotension

C. Bradycardia

D. Normal blood pressure

Answer: B

13. Symptoms and Signs of DKA

42. Severe metabolic and respiratory acidosis may manifest physically as:

A. Bradycardia

B. Tachypnea and use of accessory muscles

C. Pursed lip breathing

D. Normal respiratory rate

Answer: B

43. Which of the following is a classic "Cerebral" sign of Type II (Hypercapnic) failure?

A. Cyanosis

B. Disorientation and coma

C. Hypotension

D. Prolonged expiration

Answer: B

44. Fatigue and the inability to speak are indicators of which clinical state?

A. Effective compensation

B. Severe respiratory distress/Hypoxemia

C. Normal lung function

D. Minor airway obstruction

Answer: B

14. DIC Diagnosis: Clinical Picture and Investigations

45. Which laboratory investigation is listed as a diagnostic study to evaluate systemic response in ARF?

A. Sputum culture

B. CBC (Complete Blood Count)

C. Pulmonary function tests

D. Bronchogram

Answer: B

46. Serum electrolytes are monitored in the critically ill primarily to prevent:

A. Bronchospasm

B. Dysrhythmias

C. Alveolar collapse

D. Chest pain

Answer: B

47. Urine analysis in the diagnostic workup of the critically ill patient is used to assess:

A. Lung volumes

B. Systemic organ function and fluid status

C. Airway resistance

D. Oxygen saturation

Answer: B

48. Which diagnostic study is described as essential to confirm the diagnosis of Acute Respiratory Failure?

A. Chest radiography

B. Arterial Blood Gas (ABG)

C. Sputum culture

D. ECG

Answer: B

15. Causes of Upper GIT Bleeding

49. To prevent or treat abdominal distension, which can impair respiratory effort, which intervention is indicated?

A. Pulmonary artery catheterization

B. Insertion of a nasogastric (NG) tube

C. Artificial airway insertion

D. Pulse oximetry

Answer: B

50. Management of the critically ill patient involves preventing delivery-related complications of:

A. Only oxygen

B. Enteral or parenteral nutrition

C. Bronchodilators

D. Sedatives

Answer: B

51. Which symptom is classified as an "Extrapulmonary Symptom" in the patient's history?

A. Cough

B. Fluid retention

C. Wheeze
D. Dyspnea

Answer: B

16. Precipitants of Hepatic Encephalopathy

52. Encephalopathy in the critical care setting is often precipitated by which physiological disturbances?

A. Hypernatremia and dehydration
B. Hypoxemia, hypercapnia, and acidosis
C. Overfeeding and weight gain
D. Normal blood gas levels

Answer: B

53. When managing patients at risk for encephalopathy, clinicians must carefully assess tolerance to which class of drugs?

A. Beta2-agonists
B. Sedative, hypnotic, and narcotic drugs
C. Corticosteroids
D. Anticholinergic agents

Answer: B

54. Which underlying condition should be identified and treated to prevent further respiratory and neurological decline?

A. Sinus problems
B. Congestive heart failure
C. Sleep apnea
D. Pursed lip breathing

Answer: B

17. Nursing Interventions for Hepatic Encephalopathy

55. In patients requiring assistance with ventilation or work of breathing, which pharmacological intervention is often necessary?

A. Bronchodilators
B. Sedation
C. Antibiotics
D. Heparin

Answer: B

56. For pain control in the ICU, which class of medications should be administered?

A. Corticosteroids
B. Analgesics

C. Beta-blockers
D. Bronchodilators

Answer: B

57. Periodic assessment of a patient with neurological or respiratory distress must include:

A. Only pulse oximetry
B. Frequent Arterial Blood Gas (ABG) measurements
C. Sputum culture results
D. Daily chest radiographs

Answer: B

18. Target Organ Damage in Hypertensive Emergencies

58. Hypertensive physical findings in a respiratory patient are often secondary to:

A. Alveolar consolidation
B. Hypoxemia (Type I RF)
C. Enteral nutrition
D. Venous stasis

Answer: B

59. Which organ's distress is indicated by "Cerebral" findings like restlessness, confusion, and coma?

A. Heart
B. Brain
C. Lungs
D. Kidneys

Answer: B

60. Systemic organ function in the critically ill is monitored via which diagnostic study?

A. Sputum culture
B. Urine analysis and serum electrolytes
C. Pulmonary function tests
D. Chest radiography

Answer: B

19. Causes of ARDS

61. Acute Respiratory Distress Syndrome (ARDS) is a manifestation of which cause of respiratory failure?

A. Extrapulmonary disorders
B. Parenchymal Diseases
C. CNS disorders

D. Chest wall diseases

Answer: B

62. Severe pneumonia, a cause of ARDS, leads to hypoxemia primarily through which mechanism?

A. Alveolar hypoventilation

B. Intrapulmonary shunting

C. Decreased metabolic needs

D. Sleep apnea

Answer: B

63. Intrapulmonary shunting in ARDS occurs when blood passes through lung tissue that is:

A. Hyperventilated

B. Not ventilated (e.g., due to consolidation)

C. Over-perfused

D. Fully expanded

Answer: B

64. Alveolar consolidation, a key feature of ARDS-associated pneumonia, is a primary cause of:

A. Respiratory alkalosis

B. Intrapulmonary shunting

C. Hypernatremia

D. Bradycardia

Answer: B

20. Complications of Mechanical Ventilation

65. Venous thromboembolism (VTE) in ventilated patients is primarily precipitated by:

A. Overfeeding

B. Venous stasis resulting from immobility

C. Positive pressure ventilation

D. Beta2-agonist use

Answer: B

66. Which mechanical device is used to prevent VTE in the immobilized ICU patient?

A. Pulse oximeter

B. Intermittent pneumatic compression device

C. Nasogastric tube

D. Pulmonary artery catheter

Answer: B

67. Pharmacological prophylaxis for VTE in the ICU involves the use of:

A. Beta2-agonists

B. Unfractionated or low molecular weight heparin

C. Corticosteroids

D. Analgesics

Answer: B

68. Besides VTE, what is another category of complications mentioned in the source?

A. Complications associated with artificial airways

B. Complications associated with mechanical ventilation

C. Complications associated with enteral or parenteral nutrition

D. All of the above

Answer: D

21. Alarms of Mechanical Ventilation

69. Positive pressure is necessary in mechanical ventilation to achieve which goal?

A. To decrease cardiac output

B. To open collapsed alveoli

C. To induce hypercapnia

D. To reduce oxygen saturation

Answer: B

70. A "High Pressure" alarm on a ventilator could be triggered by which intrinsic airway cause?

A. Pulmonary emphysema

B. Large airway obstruction (e.g., foreign bodies)

C. Sleep apnea syndrome

D. Chronic bronchitis

Answer: B

71. "Positive pressure" can be delivered to a patient via which two methods?

A. NG tube or pulmonary catheter

B. Invasive artificial airway or non-invasive mask

C. Sputum culture or ABG

D. Pulse oximetry or ECG

Answer: B

72. What is a risk of delivering excessive positive airway pressure?

A. Hypokalemia

B. Oxygen toxicity or hypercapnia

C. Abdominal distension

D. Venous stasis

Answer: B

22. Eclampsia: Definition and Treatment

73. Eclampsia-related neurological dysfunction would be categorized as a cause of respiratory failure originating in the:

A. Bronchial tree

B. Central Nervous System (CNS)

C. Pleural cavity

D. Lung parenchyma

Answer: B

74. Neuromuscular findings such as muscle weakness and tremors are characteristic of which type of failure?

A. Type I RF

B. Type II RF (Hypercapnic)

C. Normal gas exchange

D. Acute Asthma

Answer: B

75. Management of a patient with CNS-related failure includes the assessment of:

A. Sputum fatigue

B. Diaphragmatic fatigue

C. Sinus problems

D. Night sweats

Answer: B

23. Management of Acute Respiratory Failure

76. The primary aims of ARF management include which of the following?

A. Promoting adequate gas exchange and correcting the underlying cause

B. Increasing $PaCO_2$ to 60 mmHg

C. Decreasing nutritional intake

D. Maintaining a pH of 7.15

Answer: A

77. Oxygenation is improved by administering supplemental oxygen through:

A. Only high-flow systems

B. Low-flow or high-flow systems

C. PFT systems

D. Bronchograms

Answer: B

78. What is the minimum targeted arterial hemoglobin oxygen saturation (SaO_2) mentioned?

A. 80%

B. 90%

C. 95%

D. 99%

Answer: B

79. If a patient's pH is less than 7.25, which intervention is specifically indicated?

A. Pursed lip breathing

B. Invasive mechanical ventilation

C. Nasogastric tube only

D. Decreasing oxygen flow

Answer: B

80. For severe respiratory failure, volume ventilation should be started in which mode?

A. Pressure Support

B. Assist/Control mode

C. CPAP

D. BiPAP

Answer: B

81. Nutrition support goals include meeting overall needs while specifically avoiding:

A. Fluids

B. Overfeeding

C. Oxygen

D. Sedation

Answer: B

82. Restoration of electrolyte balance requires prompt treatment of which two deficiencies?

A. Hypernatremia and hypercalcemia

B. Hypokalemia and hypophosphatemia

C. Hypocalcemia and hypermagnesemia

D. Normal levels

Answer: B

83. Bronchospasm in the critically ill patient should be recognized and treated with:

A. Narcotics

B. Bronchodilators and corticosteroids

C. Heparin

D. NG tubes

Answer: B

84. Which bronchodilators are mentioned as benefiting patients with airflow limitations?

A. Beta2-agonists and anticholinergic agents

B. Aspirin and Ibuprofen

C. Insulin and Glucagon

D. Heparin and Warfarin

Answer: A

85. Steroids are used in ARF management to:

A. Increase airway inflammation

B. Decrease airway inflammation and enhance beta2-agonists

C. Induce sedation

D. Prevent VTE

Answer: B

24. Complications of Acute Respiratory Failure

86. Ischemic anoxic encephalopathy is a complication resulting from:

A. Overfeeding

B. Hypoxemia, hypercapnia, and acidosis

C. High cardiac output

D. NG tube insertion

Answer: B

87. Dysrhythmias in the critically ill are precipitated by:

A. PFTs

B. Hypoxemia, acidosis, and electrolyte imbalances

C. Pursed lip breathing

D. Normal SaO_2

Answer: B

88. Prevention of VTE in the ICU involves which combination of interventions?

A. Sputum culture and chest X-ray

B. Intermittent pneumatic compression and low-dose heparin

C. NG tube and pulse oximetry

D. High-flow oxygen and sedation

Answer: B

89. To prevent abdominal distension, the nurse should perform which action?

A. Increase sedation

B. Insert a nasogastric (NG) tube

C. Administer bronchodilators
D. Monitor CVP

Answer: B

90. How is the patient's response to therapy periodically assessed?

A. Frequent ABG measurements and pulse oximetry
B. Sputum color assessment
C. Assessing for night sweats
D. Checking for sinus problems

Answer: A

25. Differential Diagnosis of Stroke

91. Conditions that are categorized alongside "Cerebrovascular accident" as CNS causes of RF include:

A. Asthma
B. Head trauma and sedative overdose
C. Chronic bronchitis
D. Pneumothorax

Answer: B

92. Cerebral symptoms in a patient history that may suggest a CNS-driven failure include:

A. Dyspnea and cough
B. Restlessness, confusion, and disorientation
C. Wheeze and chest pain
D. Sputum and night sweats

Answer: B

93. Which physical finding is a "Neuromuscular" indicator of Type II (Hypercapnic) failure?

A. Cyanosis
B. Muscle weakness and tremors
C. Hypertension
D. Prolonged expiration

Answer: B

26. Prevention of Stroke

94. A complete medical and social history is obtained to determine the patient's:

A. Future income
B. Baseline respiratory status on admission
C. Sputum volume

D. Cardiac output

Answer: B

95. Which extrapulmonary symptoms should be profiled in the history to assess the risk of respiratory decline?

A. Dyspnea and cough

B. Night sweats, headache on awakening, and fatigue

C. Wheeze and chest pain

D. Sputum color and consistency

Answer: B

96. The "Previous History" assessment includes which of the following?

A. Pulse oximetry values

B. Medical diagnosis, treatments, and immunization

C. Arterial blood pH

D. Current $PaCO_2$ levels

Answer: B

27. Management of Ischemic and Hemorrhagic Stroke

97. The primary goal of managing ARF caused by CNS disorders like stroke is to:

A. Increase CO_2 levels

B. Correct the underlying cause and promote gas exchange

C. Decrease oxygen delivery

D. Induce metabolic acidosis

Answer: B

98. When managing a patient with a CNS disorder, why is it critical to assess tolerance to sedative and narcotic drugs?

A. Because they increase airway inflammation

B. Because they can further depress respiratory drive

C. Because they cause hypokalemia

D. Because they lead to VTE

Answer: B

99. If diaphragmatic fatigue is identified in a patient with a CNS injury, what is the appropriate action?

A. Decrease nutrition

B. Initiate/maintain mechanical ventilation support

C. Administer only bronchodilators

D. Perform a bronchogram

Answer: B

100. Which intervention is indicated for the specific complication of pleural effusion or pneumothorax?

A. Insertion of a nasogastric tube

B. Removal of air or fluid in the pleural cavity

C. Administration of beta2-agonists

D. Increasing sedative dosages

Answer: B